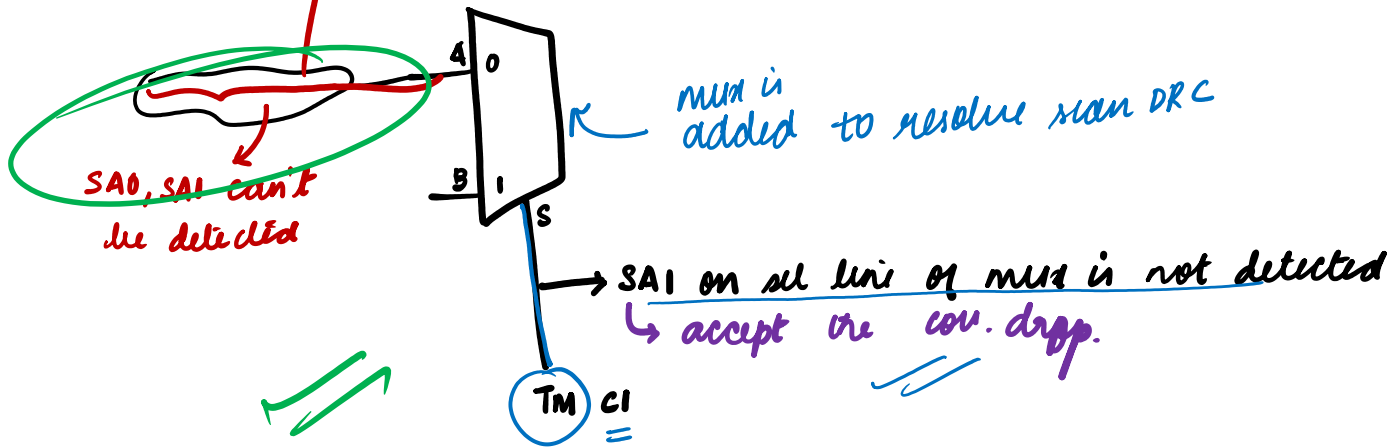
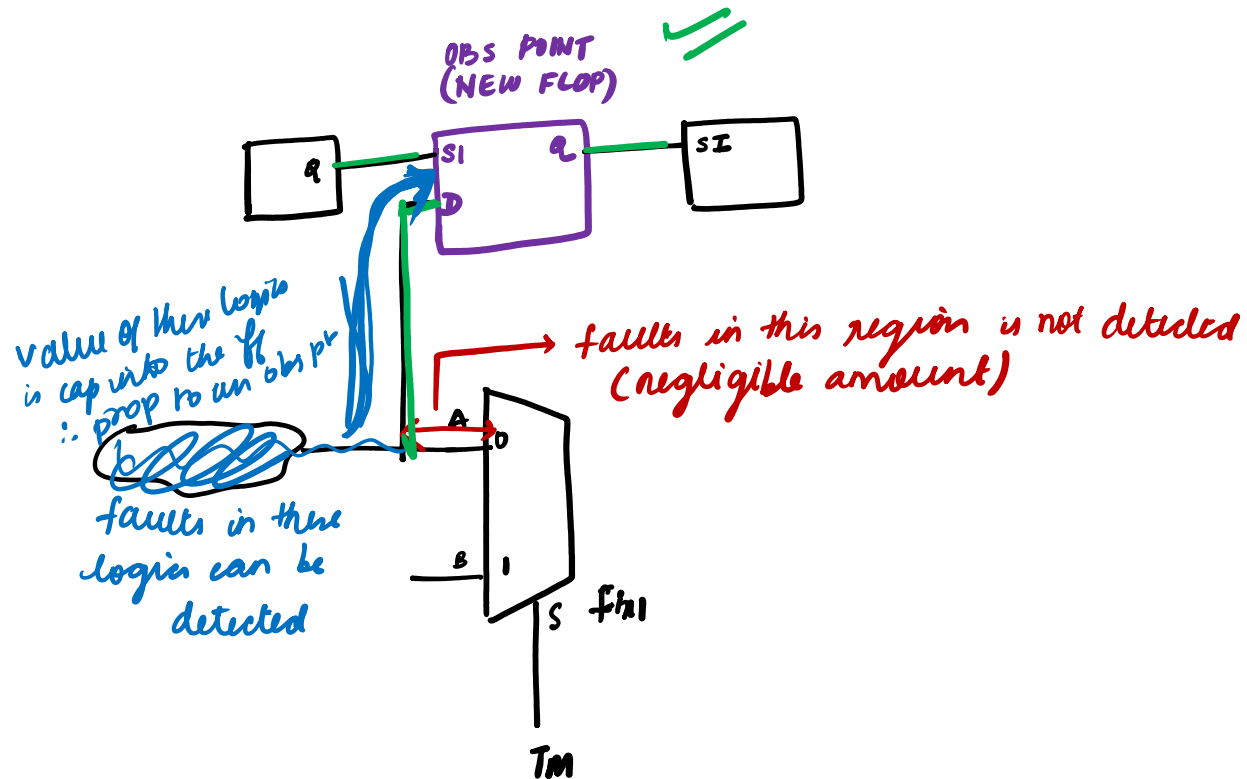


Q: Some situations which resulted in cov drop and how did you improve?

(i) AU.PC :- mux to resolved scan DRC.
sel line of mux connected TM CI $\rightarrow$ TM CI $\rightarrow$ TM = 1 throughout DFT.
 $\therefore$ A i/p (0th i/p) of mux is never sel in DFT. So the value of the logics connected to 0th i/p of mux can't be prop to oth pt. $\therefore$ SA0, SA1 on these logics $\Rightarrow$ AU-PC.
 $\rightarrow$ Since A i/p of mux can't be selected during DFT mode, the value of these logics can't be propagated to an oth pt.



Improving the coverage :-



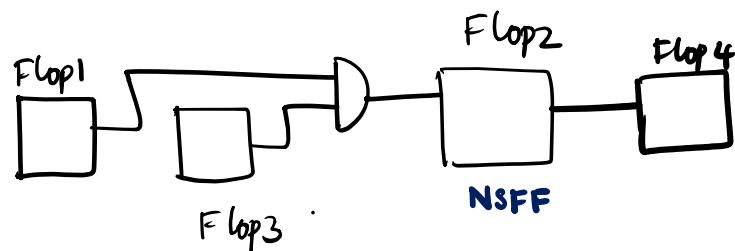
② AU. SEQ $\Rightarrow$ Non scan FF b/w 2 scan FFs.

NO TE

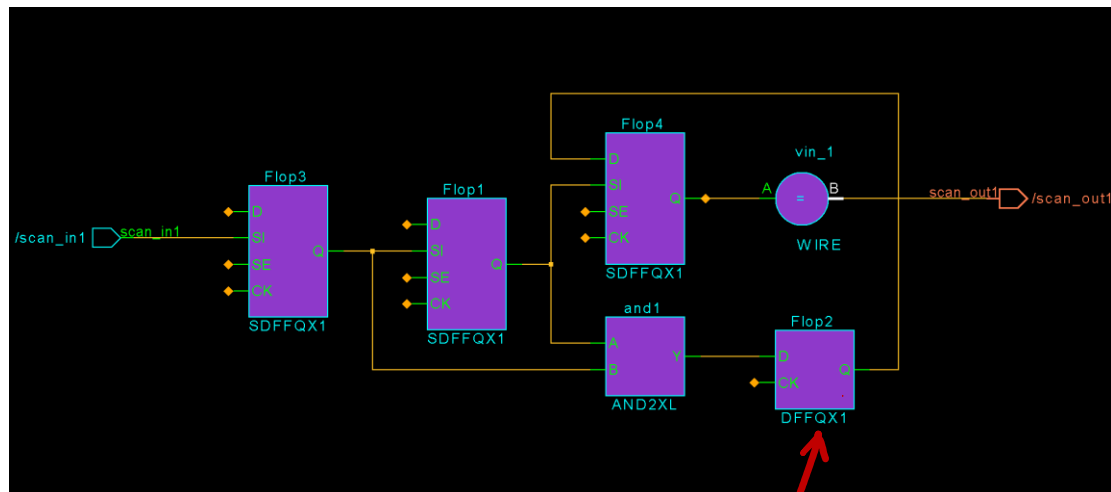
رأى

In which situations, by increasing req depth, we can improve the coverage.

Ans:- When there is non scan ff b/w 2 SFFs (theory class concept)



← Design before scan insertion



← Design after scan insertion

NSFF

ATPG :-

//Setup phase

set_context patterns -scan

read_verilog /hdd2/home/vidishar/interview_batch/seq_depth/scan/outputs/try_scan.v -verbose

read_cell_library /hdd2/home/vidishar/nov24/library/tsmc13.mdt

set_current_design

dofile /hdd2/home/vidishar/interview_batch/seq_depth/scan/outputs/atpg.dofile

//Analysis Phase

set_system_mode analysis

set_fault_type stuck

add_faults -all

set_pattern_type -sequential 1

create_patterns

one clk pulse in cap

COVERAGE :

Coverage

test_coverage

fault_coverage

73.21%

73.21%

Covimp: increase the seq depth

//Setup phase

set_context patterns -scan

read_verilog /hdd2/home/vidishar/interview_batch/seq_depth/scan/outputs/try_scan.v -verbose

read_cell_library /hdd2/home/vidishar/nov24/library/tsmc13.mdt

set_current_design

dofile /hdd2/home/vidishar/interview_batch/seq_depth/scan/outputs/atpg.dofile

//Analysis Phase

set_system_mode analysis

set_fault_type stuck

add_faults -all

set_pattern_type -sequential 2

upto 2 clk pulses in cap

Coverage

test_coverage

fault_coverage

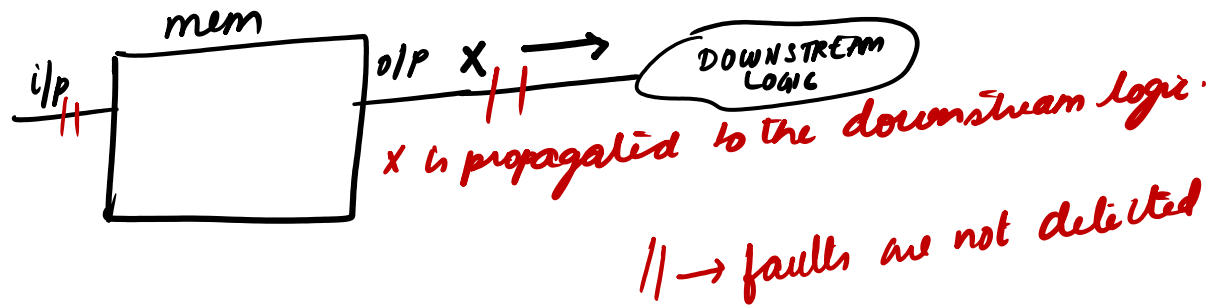
100.00%

100.00%

Command →

(iii)

Memories are there in the design.



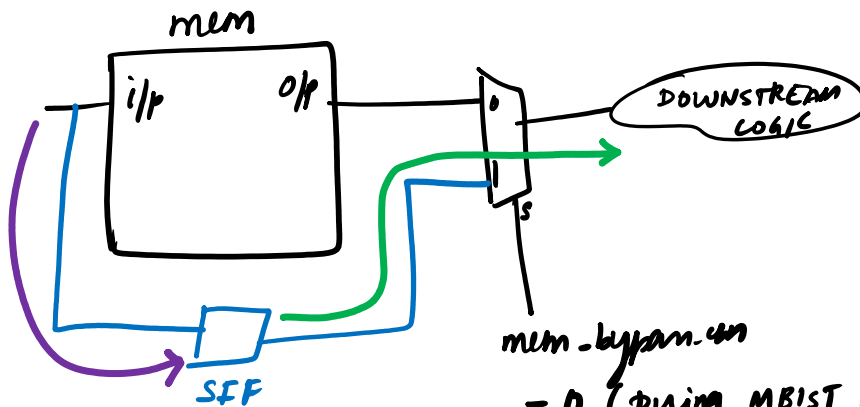
Cov inup

Soln:- Bypass the mem to stop the prop of α to the downstream logic

SFF:-

☐ IL will act like an obs pt for the fault sites at the i/p of the mem.

(i.e., the value of the i/p of mem cap will be into that fl) [repeated by violet line] See #



mem-bypass-en
= 0 (during MBIST or In mode)
= 1 (otherwise)

* It will also act as a control fl for the downstream logic.
(i.e., the last shifted value of the fl will control the downstream logic)
[repeated by green colour arrow]

(iv)

Undetected faults (u_0, u_c)

We can improve the coverage by adding testpoints

To cover the faults inside u_0, u_c class, there is an automatic flow of adding testpoints using Tesent tool.

TESTPOINT INSERTION SCRIPT:

```
1 set_context dft -test_points -no_rtl
2 read_verilog case6.v
3 read_cell_library /hdd2/home/vidishar/nov24/library/tsmc13.mdt
4 set_current_design
5
6 add_clocks 0 BIST_CLK
7 add_clocks 1 MBIST_RST_L
8
9 set_test_point_type atpg
10 set_system_mode analysis
11 set_test_point_analysis_options -total_number 100
12 analyze_test_points
13 insert_test_logic
14 write_design -output_file updated.v -rep
--
```

Specifies the max no. of testpoints to be created.

Specifies the testpoint analysis and generates the list of testpoints to be inserted by tool.

orig design (case6.v)
+
new TPs added.

Inserts the TPs

Commands to do the TPI.

(v)

UC+U0	(pen_constraints)	1 (0.00%)	same (0.00%)
EAB	(edt_abort)	726 (3.33%)	same (3.34%)

- **UD.EAB — EDT_ABORT**

If our design's chain:channel ratio is insufficient to detect the fault.

↳ You can use `analyze_compression` command to determine the compression ratio of that particular project.

analyze-compression :

Analyzes a scan inserted design and returns a range of maximum chain:channel ratio

where the test cov begins to decline.

→ if your comp ratio is 2x, then from there your test cov begins to decline.

```
// For stuck-at faults
//
// Chain:Channel ratio Predicted fault coverage drop
// -----
// 2 negligible fault coverage drop
// 3 0.01% - 1.00% drop
// CPU time to analyze compression is 0 seconds.
```

we are not using
∴ 1x compression → no. of int ch = no. of ext ch
∴ no point in going for comp

→ Comp ratio is 1x ⇒ 100% Tc or almost equal to 100% TC.

Comp ratio is 2x ⇒ Tc will decrease slightly. → no. of int chain = 4
no. of ext ch = 2. cov = 99.98% Preferred config

Comp ratio is 3x ⇒ Tc will decrease further → no. of int chain = 6
no. of ext ch = 2 cov = 98.83%
→ comp ratio 3x and above ⇒ cov drops below 99%.